

Westravelle Analytical Ltd

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Chamber 04 Temperature and Humidity Mapping Certificate

ICH chamber programme, post-installation mapping

Certificate no. E-M-2018-004

Date: 14/09/2018

Mapping protocol: MP-CH04-01, revision 2

1 Purpose

This certificate records the temperature and relative humidity mapping of chamber 04 in the ICH chamber programme after the unit was moved to the Westfield Road laboratory and recommissioned. It confirms that the chamber holds its set points across the working volume before any stability pull is placed in it.

2 Chamber and set points

Chamber 04 was mapped at the 25 °C/60% RH and 40 °C/75% RH set points. The 30 °C/65% RH set point was not mapped for this certificate because it is not used for stability pulls until the next scheduled review.

Temperature and relative humidity were recorded with calibrated data loggers placed at the logger positions defined in the mapping protocol, covering the upper, middle and lower shelves at the front, centre and rear of the chamber, plus one logger at the door seal.

3 Acceptance criteria

Per protocol MP-CH04-01, the mapping passes when every logger stays within the set point tolerance for the full stabilised period, with no single reading outside the tolerance band after the first hour of the run. Spread between loggers is reported against the protocol limit and is not a separate pass criterion.

- Temperature within the set point tolerance at all logger positions
- Relative humidity within the set point tolerance at all logger positions
- Door-seal logger reviewed against the centre logger for excursions on opening

4 Results

All loggers met the acceptance criteria at both set points. The door-seal logger showed a short rise in humidity during the simulated door-opening step, recovering within the protocol recovery time. This is recorded as an observation and does not affect the pass result.

The mapping run was repeated once at 40 °C/75% RH after the first run was voided for a logger connection fault. The voided run is kept in the chamber file with the reason written against it.

5 Deviations

One deviation was raised against the logger record (see deviation note DN-2018-031). The serial number of the humidity probe at the rear upper position was entered incorrectly on the logger record sheet and corrected by hand on the original sheet, initialled and dated. The corrected serial number has been checked against the calibration certificate on file.

6 Conclusion

Chamber 04 is accepted for stability pulls at 25 °C/60% RH and 40 °C/75% RH from the date of this certificate. Re-mapping is due after any change to the refrigeration or humidification system, or after 12 months, whichever is sooner.

Name	Role	Signature	Date
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Appendix A Logger calibration reference

Calibration certificates for the loggers used in this mapping are filed in the instrument folder for chamber 04. Certificate numbers are recorded on the logger record sheet.